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High-frequency trading in the stock market and the costs of options market making

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1 Big picture

- **Question.** How does high-frequency trading (HFT) in the underlying stock market affect liquidity in the options market?
- **Setting.** The paper links stock-market HFT activity to options-market bid–ask spreads for U.S. stocks and options, focusing on 2009 NASDAQ HFT data and OPRA options data.
- **Core result.** More HFT in the stock market is associated with wider options bid–ask spreads. The effect is driven by **liquidity-demanding/aggressive HFT**, not liquidity-supplying HFT.
- **Economic magnitude.** A one-standard-deviation increase in stock-market HFT activity is associated with about a 3.5% increase in proportional options spreads. For 1,000 option contracts, trading costs rise from roughly \$85.91 to \$88.92.
- **Channel 1: latency arbitrage.** Aggressive HFTs exploit cross-market mispricings when options quotes are stale relative to the stock. Options market makers then widen spreads to protect themselves from being sniped.
- **Channel 2: informed trading.** Options attract informed traders because of embedded leverage. Informed trading in options increases options adverse selection and can simultaneously create stock-market HFT opportunities.
- **Cross-sectional evidence.** The HFT effect is stronger for stocks with more profitable put–call parity violations, which is consistent with cross-market latency arbitrage.
- **Event evidence.** The paper uses the arrest of Raj Rajaratnam as an exogenous shock to informed trading. The informed-trading channel is stronger before the arrest.
- **IV evidence.** To address endogeneity, the paper uses NASDAQ flash orders as an instrument for HFT activity. The 2SLS estimates confirm that aggressive HFT increases options spreads.
- **Takeaway.** Speed in one market can raise trading costs in a linked market. Stock-market HFT does not only affect stock liquidity; it also changes the cost of market making in options.

2 Data and measurement

2.1 Primary sample

- The primary sample uses NASDAQ HFT data and OPRA options transactions data.
- The main period is 2009, which is the year for which the detailed NASDAQ HFT data are available.
- The main analysis covers 103 underlying stocks and their associated options.
- The main panel is at the stock-day level.
- Table 2 reports 20,639 observations for the main stock-day variables.

Interpretation: The sample is narrow but unusually rich: it distinguishes liquidity-supplying from liquidity-demanding HFT. That distinction is the key to the paper because the effects are concentrated in aggressive HFT.

2.2 Main outcome: options-market bid–ask spread

The main dependent variable is the proportional options bid–ask spread:

$$OPspread_{i,d}.$$

It is computed for stock i on day d as the volume-weighted average proportional spread across option transactions:

$$OPspread = \frac{Ask - Bid}{(Ask + Bid)/2}.$$

Interpretation: Options spreads measure market makers' compensation for adverse selection, inventory risk, hedging costs, and order processing costs. Wider spreads mean higher trading costs for options investors.

2.3 HFT variables

The main HFT variables are:

- $SHFT_{i,d}^{All}$: total HFT trading volume in the underlying stock;
- $SHFT_{i,d}^S$: liquidity-supplying HFT volume;
- $SHFT_{i,d}^D$: liquidity-demanding HFT volume;
- $SNHFT_{i,d}^S$ and $SNHFT_{i,d}^D$: corresponding non-HFT supply and demand trading volumes.

Interpretation: The distinction between $SHFT^S$ and $SHFT^D$ matters because the paper's mechanism is not simply "more electronic trading." It is aggressive, liquidity-consuming speed that raises options market-making costs.

2.4 Options controls and stock controls

The controls include variables from both the options and underlying stock markets:

- option volume $Ovolume_{i,d}$;
- option implied volatility $Oimplied_{i,d}$;
- inverse option price $Oinp_{i,d}$;
- absolute delta $|Odelta_{i,d}|$;
- gamma $Ogamma_{i,d}$;
- vega $Ovega_{i,d}$;
- stock spread $SPspread_{i,d}$;
- stock volatility $SVolatility_{i,d}$.

Interpretation: These variables absorb standard option market-making costs: hedging demand, option moneyness, volatility, inventory risk, and stock-market trading frictions.

2.5 Latency-arbitrage proxy

The paper measures profitable put–call parity violations. The option-implied stock price is

$$S^* = C + Xe^{-(r-q)T} - P,$$

where C is the call price, P is the put price, X is the strike, r is the risk-free rate, q is the dividend yield, and T is time to maturity.

A profitable arbitrage opportunity exists if the option-implied stock price differs from the actual stock price enough to overcome transaction costs. The paper counts the number of profitable violations:

$$Npv_{i,d}.$$

It then defines $High_{i,d} = 1$ when $Npv_{i,d}$ is above the daily median.

Interpretation: If aggressive HFTs raise options spreads by sniping stale quotes, the effect should be stronger for stocks with more profitable put–call parity violations.

2.6 Informed-trading proxy

The informed-trading measure uses CBOE open-buy put and call volume:

$$Ins_{i,d} = \left| \frac{OBPV}{OBPV + OBCV} - 0.5 \right|,$$

where $OBPV$ is open-buy put volume and $OBCV$ is open-buy call volume.

Interpretation: The absolute deviation from 0.5 measures directional concentration in opening option trades. A large value indicates more informed trading, regardless of whether the information is positive or negative.

3 Models and empirical specifications

3.1 Baseline OLS specification

The baseline specification is

$$OPspread_{i,d} = \alpha_i + \beta_d + \gamma_1 SHFT_{i,d}^{All} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}. \quad (1)$$

The paper also estimates separate supply and demand HFT specifications:

$$OPspread_{i,d} = \alpha_i + \beta_d + \gamma_1 SHFT_{i,d}^S + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}, \quad (2)$$

$$OPspread_{i,d} = \alpha_i + \beta_d + \gamma_1 SHFT_{i,d}^D + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}. \quad (3)$$

Interpretation: The fixed effects absorb stock-level differences and common daily shocks. The coefficient γ_1 asks whether stock-market HFT is associated with higher options spreads after controlling for standard market-making costs.

3.2 Latency-arbitrage channel specification

The latency-arbitrage test interacts HFT with the high put–call parity violation indicator:

$$OPspread_{i,d} = \alpha_i + \beta_d + \gamma_1 SHFT_{i,d} + \gamma_2 High_{i,d} + \gamma_3 SHFT_{i,d} High_{i,d} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}. \quad (4)$$

Interpretation: The key coefficient is γ_3 . A positive γ_3 means HFT raises options spreads more when there are more exploitable cross-market arbitrage opportunities.

3.3 Informed-trading channel specification

Around the arrest of Raj Rajaratnam, the paper estimates

$$OPspread_{i,d} = \alpha_i + \gamma_1 Ins_{i,d-1} + \gamma_2 Event_{i,d} + \gamma_3 Ins_{i,d-1} Event_{i,d} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}, \quad (5)$$

where $Event_{i,d} = 1$ before the arrest and zero after it.

It also estimates HFT activity as an outcome:

$$SHFT_{i,d} = \alpha_i + \gamma_1 Ins_{i,d-1} + \gamma_2 Event_{i,d} + \gamma_3 Ins_{i,d-1} Event_{i,d} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}. \quad (6)$$

Interpretation: If informed trading is a channel, $Ins \times Event$ should be positive before the arrest: informed options trading should predict wider options spreads and more aggressive HFT activity in the underlying stock.

3.4 Joint informed-trading and HFT specification

The paper then estimates

$$\begin{aligned} OPspread_{i,d} = & \alpha_i + \gamma_1 Ins_{i,d-1} + \gamma_2 Event_{i,d} + \gamma_3 Ins_{i,d-1} Event_{i,d} \\ & + \gamma_4 SHFT_{i,d}^D + \gamma_5 SHFT_{i,d}^D Event_{i,d} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}. \end{aligned} \quad (7)$$

Interpretation: This asks whether informed trading fully explains the HFT-spread relation. The paper finds it does not. Liquidity-demanding HFT remains important, but informed trading amplifies its effect by roughly 50%.

3.5 2SLS IV specification

To address endogeneity, the paper instruments HFT with the introduction of NASDAQ flash orders. Conceptually, the first stage is

$$SHFT_{i,d} = \alpha_i + \beta_d + \pi IV_{i,d} + \sum_k \delta_k C_{k,i,d} + u_{i,d}, \quad (8)$$

where $IV_{i,d}$ captures periods around the flash-order regime.

The second stage is

$$OPspread_{i,d} = \alpha_i + \beta_d + \gamma \widehat{SHFT}_{i,d} + \sum_k \delta_k C_{k,i,d} + \epsilon_{i,d}. \quad (9)$$

Interpretation: Flash orders affect fast traders because only low-latency firms can observe and respond within 500 milliseconds. The exclusion restriction is that flash orders affect options spreads through their effect on stock-market HFT.

4 Identification and empirical design

4.1 Main empirical challenge

- Options and stocks are intrinsically linked.
- HFT may respond to the same information that changes options spreads.
- Market-wide shocks can affect both stock HFT and options liquidity.
- Options market liquidity and stock-market HFT may be jointly determined.

Interpretation: The OLS results are informative but potentially endogenous. This is why the paper uses channel tests and IV tests.

4.2 Fixed effects and controls

- Stock fixed effects absorb persistent differences across firms.
- Day fixed effects absorb aggregate market conditions.
- Options and stock controls absorb hedging costs and option-specific liquidity determinants.
- Standard errors are double clustered by firm and day.

Interpretation: This design compares changes in HFT and options spreads within the same stock and across the same trading days.

4.3 Latency-arbitrage identification

- Put-call parity links option prices and the underlying stock price.
- When option quotes are stale, fast arbitrageurs can trade against slow market makers.
- Stocks with more profitable put-call violations should expose options market makers to more sniping risk.

Interpretation: The interaction with $High_{i,d}$ is a cross-sectional mechanism test: aggressive HFT should matter more when the arbitrage opportunity set is larger.

4.4 Informed-trading identification

- The Rajaratnam arrest is used as a shock that reduces informed trading in options.
- The paper tests whether informed-trading measures predict spreads and aggressive HFT before the arrest.
- It also checks whether the HFT-spread link is stronger before the arrest.

Interpretation: This test separates the informed-trading channel from the latency-arbitrage channel. The two channels are complementary, not mutually exclusive.

4.5 IV identification

- NASDAQ flash orders provide an exogenous shock to HFT activity.
- Flash orders expose marketable orders for 500 milliseconds before routing to the market.
- Low-latency traders are best positioned to react to these orders.
- The second-stage regressions use fitted HFT activity to estimate its impact on options spreads.

Interpretation: The IV evidence supports a causal interpretation: an increase in aggressive HFT in the stock market raises options-market trading costs.

5 Tables and figures

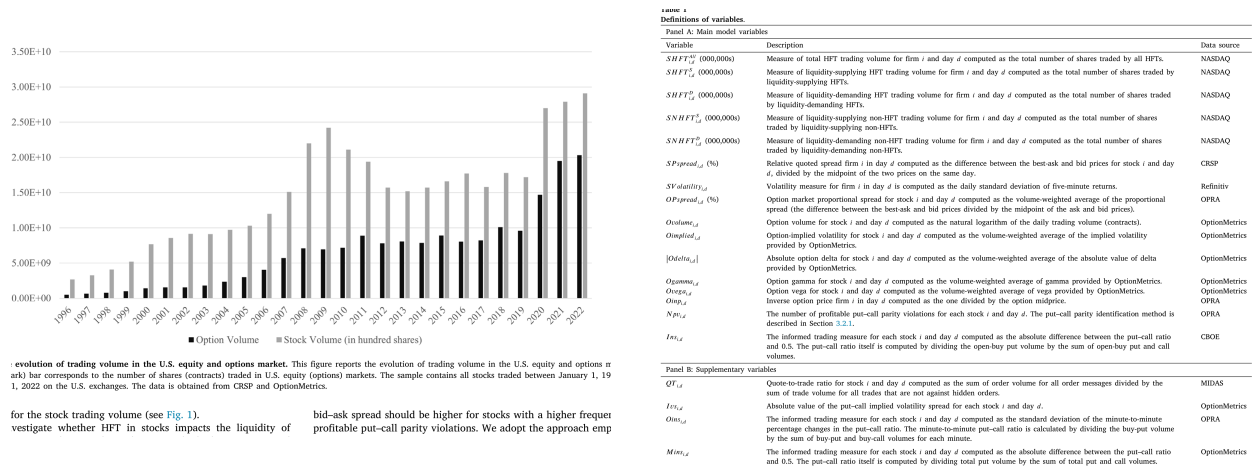
The visuals below are directly cropped from the original paper and grouped to save space.

5.1 Figure 1 and Table 1: market growth and variable definitions

Read:

- Figure 1 shows that U.S. options trading volume grew substantially from 1996 to 2022.
- Options volume grows faster than stock trading volume, which motivates studying the cost of options liquidity.
- Table 1 defines the stock HFT variables, options liquidity variables, options Greeks, put–call parity violation measures, and informed-trading measures.

Economic message: The paper studies an economically important market. Options volume has become large enough that small changes in spreads imply large aggregate trading-cost effects.



5.2 Tables 2–3: summary statistics and baseline HFT effect

Read: Table 2:

- Average total HFT stock trading volume is 1.36 million shares per stock-day.

- Average liquidity-supplying HFT volume is 0.93 million shares, and average liquidity-demanding HFT volume is 0.77 million shares.
- The mean proportional options spread is 21.24%, while the mean stock spread is only 0.12%.
- The average stock-day has 4.49 profitable put–call parity violations.

Read: Table 3:

- Total HFT has a positive and significant effect on options spreads: coefficient 0.03 with a t-statistic of 2.66.
- Liquidity-supplying HFT has no significant effect.
- Liquidity-demanding HFT has a positive and significant effect: coefficient 0.03 with a t-statistic of 3.11.

Economic message: The baseline result is clear: the spread effect comes from aggressive HFT, not passive HFT supply.

Table 2

Summary statistics.

Panel A: Main model variables				
	Mean	Median	Std. dev.	N
$SHFT_{i,d}^{All}$ (000,000s)	1.36	0.20	2.81	20,639
$SHFT_{i,d}^S$ (000,000s)	0.93	0.10	2.11	20,639
$SHFT_{i,d}^D$ (000,000s)	0.77	0.14	1.48	20,639
$SNHFT_{i,d}^S$ (000,000s)	0.98	0.27	1.67	20,639
$SNHFT_{i,d}^D$ (000,000s)	1.13	0.23	2.27	20,639
$SPspread_{i,d}$ (%)	0.12	0.08	0.12	20,639
$SVolatility_{i,d}$	0.003	0.003	0.002	20,639
$OPspread_{i,d}$ (%)	21.24	13.09	24.77	20,639
$Ovolume_{i,d}$	6.80	6.80	3.16	20,639
$Oimplied_{i,d}$	0.54	0.48	0.23	20,639
$[Odelta]_{i,d}$	0.52	0.52	0.13	20,639
$Ogamma_{i,d}$	0.10	0.08	0.06	20,639
$Ovega_{i,d}$	4.91	3.76	4.08	20,639
$Oimp_{i,d}$	1.27	0.91	1.22	20,639
$Npv_{i,d}$	4.49	0.00	11.83	8344
$Ins_{i,d}$	0.25	0.24	0.20	4892
Panel B: Supplementary variables				
$QT_{i,d}$	42.07	33.73	29.43	1,862,924
$Ivs_{i,d}$	0.04	0.02	0.06	1,862,924
$Oins_{i,d}$	0.22	0.01	0.94	18,994
$Mins_{i,d}$	0.46	0.25	0.76	1,862,924

This table reports the summary statistics (mean, median, and standard deviation) of the

(a) Table 2: summary statistics.

Table 3

The impact of HFT activities on option spread – OLS.

	$OPspread_{i,d}$ (i)	$OPspread_{i,d}$ (ii)	$OPspread_{i,d}$ (iii)
$SHFT_{i,d}^{All}$	0.03*** (2.66)		
$SHFT_{i,d}^S$		0.01 (0.73)	
$SHFT_{i,d}^D$			0.03*** (3.11)
Controls	Yes	Yes	Yes
Stock fixed effect	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes
N obs.	20,639	20,639	20,639
R ²	0.38	0.38	0.38

This table reports the results of estimating the impact of HFT in equity markets on the options bid–ask spread using the following models:

$$OPspread_{i,d} = \alpha_i + \beta_d + \gamma_1 SHFT_{i,d}^{All} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}$$

$$OPspread_{i,d} = \alpha_i + \beta_d + \gamma_1 SHFT_{i,d}^S + \sum_{k=1}^9 \delta_k C_{k,i,d} + \epsilon_{i,d}$$

$$OPspread_{i,d} = \alpha_i + \beta_d + \gamma_1 SHFT_{i,d}^D + \sum_{k=1}^9 \delta_k C_{k,i,d} + \epsilon_{i,d}$$

where $OPspread_{i,d}$ is the proportional option bid–ask spread, $SHFT_{i,d}^{All}$ is the measure of HFTs' total activities, $SHFT_{i,d}^S$ is the measure of HFTs' liquidity-supplying activities, and $SHFT_{i,d}^D$ is the measure of HFTs' liquidity-demanding activities. $C_{k,i,d}$ is a set of k control variables including variables from both the options and underlying markets

(b) Table 3: baseline OLS.

5.3 Table 4: latency-arbitrage channel

Read:

- For total HFT, the interaction $SHFT^{All} \times High$ is 0.06 and statistically significant.
- For liquidity-demanding HFT, the interaction $SHFT^D \times High$ is 0.07 and statistically significant.
- The main HFT coefficients become negative or insignificant once the interaction is included, showing that the spread effect is concentrated in high-arbitrage-opportunity stocks.

Economic message: Options market makers widen spreads especially when stock-market HFT can exploit stale options quotes through put–call parity arbitrage.

Table 4
Testing the latency arbitrage channel – OLS.

	$OPspread_{i,d}$ (i)	$OPspread_{i,d}$ (ii)
$SHFT_{i,d}^{All}$	-0.06* (-1.89)	
$SHFT_{i,d}^D$		-0.05 (-1.40)
$HIGH_{i,d}$	-0.05*** (-4.43)	-0.06*** (-4.45)
$SHFT_{i,d}^{All} HIGH_{i,d}$	0.06*** (3.67)	
$SHFT_{i,d}^D HIGH_{i,d}$		0.07*** (3.59)
Controls	Yes	Yes
Stock fixed effect	Yes	Yes
Time fixed effect	Yes	Yes
N obs.	8344	8344
R^2	0.28	0.28

This table reports the results of testing the latency arbitrage channel using the following models:

$OPspread_{i,d} = \alpha_i + \beta_d + \gamma_1 SHFT_{i,d}^{All} + \gamma_2 HIGH_{i,d} + \gamma_3 SHFT_{i,d}^{All} HIGH_{i,d} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}$
 $OPspread_{i,d} = \alpha_i + \beta_d + \gamma_1 SHFT_{i,d}^D + \gamma_2 HIGH_{i,d} + \gamma_3 SHFT_{i,d}^D HIGH_{i,d} + \sum_{k=1}^9 \delta_k C_{k,i,d} + \epsilon_{i,d}$
where $OPspread_{i,d}$ is the proportional option bid-ask spread, $SHFT_{i,d}^{All}$ is the measure of HFTs' total activities and $SHFT_{i,d}^D$ is the measure of HFTs' liquidity-demanding activities. $HIGH_{i,d}$ is a dummy variable equal to one in day d if its $Npv_{i,d}$ (the number of put-call parity violations) value is above the median number of profitable put-call parity violations for all stocks. $C_{k,i,d}$ is a set of k control variables, including variables from both the options and underlying markets. The options market variables are $Ovolume_{i,d}$, $Oimplied_{i,d}$, $Oinp_{i,d}$, $|Odelta_{i,d}|$, $Ogamma_{i,d}$, and $Ovega_{i,d}$, and the stock market variables are $SPspread_{i,d}$, and $SVolatility_{i,d}$. In addition to these variables, we also control for non-HFTs' liquidity-demanding activities ($SNHFT_{i,d}^D$) when we use $SHFT_{i,d}^D$. All models include both firm and day fixed effects (α_i and β_d respectively). The standard errors used to compute the t-statistics (in brackets) are double clustered by firm and day. *, **, and *** denote the significance at 10%, 5%, and 1% respectively. For detailed definitions of variables refer to Table 1.

Table 4: latency-arbitrage channel.

5.4 Table 5, Figure 2, and Table 6: informed-trading channel

Read: Table 5:

- The interaction $Ins_{i,d-1} \times Event_{i,d}$ is positive and significant for options spreads.
- The same interaction is positive for liquidity-demanding HFT, but not for liquidity-supplying HFT.
- This means informed options activity predicts both wider options spreads and more aggressive stock-market HFT before Rajaratnam's arrest.

Read: Figure 2:

- The lag-one informed-trading signal has the strongest predictive effect for option spreads.
- The confidence intervals become wider and less informative at longer lags.

Read: Table 6:

- $SHFT^D$ remains positive and significant after controlling for informed trading.
- $SHFT^D \times Event$ is also positive and significant.
- The paper interprets this as informed trading amplifying the aggressive-HFT effect by roughly 50%.

Economic message: Informed trading does not fully explain the HFT effect, but it strengthens it. Options information can trigger aggressive stock-market HFT and raise market-making costs in options.

Table 5

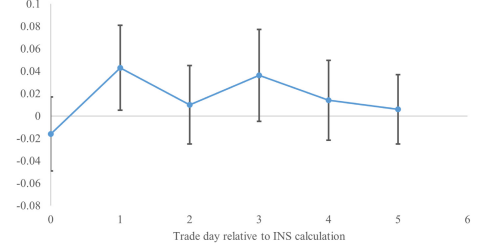
Testing the informed trading channel – main analysis (part I)

	$OPspread_{i,d}$ (i)	$OPspread_{i,d}$ (ii)	$SHFT_{i,d}^S$ (iii)	$SHFT_{i,d}^S$ (iv)	$SHFT_{i,d}^D$ (v)	$SHFT_{i,d}^D$ (vi)
$Ins_{i,d-1}$	-0.01 (-0.67)	-0.04** (-2.33)	-0.01 (-0.87)	-0.01* (-1.72)	0.00 (0.23)	-0.01 (-1.30)
$Event_{i,d}$		-0.03 (-1.41)		-0.01 (-0.82)		-0.05** (-2.27)
$Ins_{i,d-1}Event_{i,d}$		0.04** (2.43)		0.01 (0.84)		0.02** (1.96)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Stock fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effect	Yes	No	Yes	No	Yes	No
N obs.	4892	4892	4892	4892	4892	4892
R^2	0.38	0.39	0.03	0.06	0.03	0.08

This table reports the results of testing the informed trading channel using the following models:

$OPspread_{i,d} = \alpha_i + \gamma_1 Ins_{i,d-1} + \gamma_2 Event_{i,d} + \gamma_3 Ins_{i,d-1}Event_{i,d} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}$
 $SHFT_{i,d}^S = \alpha_i + \gamma_1 Ins_{i,d-1} + \gamma_2 Event_{i,d} + \gamma_3 Ins_{i,d-1}Event_{i,d} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}$
where $OPspread_{i,d}$ is the proportional option bid-ask spread, $Ins_{i,d}$ is the measure of informed trading, and $SHFT_{i,d}$ is one of two HFT measures ($SHFT_{i,d}^S$ and $SHFT_{i,d}^D$). $Event_{i,d}$ is a dummy variable equal to one before October 16, 2009 (Rajaratnam's arrest) and zero thereafter. The sample encompasses a period of two months before and after the event date, with the exclusion of the five trading days preceding the event date. $C_{k,i,d}$ is a set of k control variables, including variables from both the options and underlying markets. The options market variables are $Ovolume_{i,d}$, $Oimplied_{i,d}$, $Oinpi_{i,d}$, $|Odelta_{i,d}|$, $Ogamma_{i,d}$, and $Ovega_{i,d}$, and the stock market variables are $SPspread_{i,d}$ and $SVolatility_{i,d}$. In columns (i), (iii), and (v), both firm and day fixed effects (α_i and β_j respectively) are included, while columns (ii), (iv), and (vi) include only firm fixed effects. The standard errors used to compute the t-statistics (in brackets) are double clustered by firm and day. *, **, and *** denote

(a) Table 5: informed trading, part I.



lity of open-buy option volume signal for options bid-ask spread. This figure reports the coefficient and the 95% confidence intervals of the open-buy option volume signal for options bid-ask spread at various lags (0, -1-day, -2-day, -3-day, -4-day, and -5-day) before the arrest of Raj Rajaratnam.

(b) Figure 2: lag structure.

Table 6

Testing the informed trading channel – main analysis (part II)

	$OPspread_{i,d}$
$Ins_{i,d-1}$	-0.04** (-2.31)
$Event_{i,d}$	-0.04 (-1.55)
$Ins_{i,d-1}Event_{i,d}$	0.04** (2.40)
$SHFT_{i,d}^D$	0.04*** (5.52)
$SHFT_{i,d}^D Event_{i,d}$	0.02** (2.03)
Controls	Yes
Stock fixed effect	Yes
Time fixed effect	No
N obs.	4892
R^2	0.40

This table reports the results of testing the informed trading channel using the following model:

$$OPspread_{i,d} = \alpha_i + \gamma_1 Ins_{i,d-1} + \gamma_2 Event_{i,d} + \gamma_3 Ins_{i,d-1}Event_{i,d} + \gamma_4 SHFT_{i,d}^D + \gamma_5 SHFT_{i,d}^D Event_{i,d} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}$$

where $OPspread_{i,d}$ is the proportional option bid-ask spread, $SHFT_{i,d}^D$ is the measure of HFTs' liquidity-demanding activities, and $Ins_{i,d}$ is the measure of informed trading. $Event_{i,d}$ is a dummy variable equal to one before October 16, 2009 (Rajaratnam's arrest) and zero thereafter. The sample encompasses a period of two months before and after the event date, with the exclusion of the five trading days preceding the event date. $C_{k,i,d}$ is a set of k control variables, including variables from both the options and underlying markets. The options market variables are $Ovolume_{i,d}$, $Oimplied_{i,d}$, $Oinpi_{i,d}$, $|Odelta_{i,d}|$, $Ogamma_{i,d}$, and $Ovega_{i,d}$, and the stock market variables are $SPspread_{i,d}$ and $SVolatility_{i,d}$. The table includes only firm fixed effects. The standard errors used to compute the t-statistics (in brackets) are double clustered by firm and day. *, **, and *** denote the significance at 10%, 5%, and 1% respectively. For detailed definitions of variables refer to Table 1.

Table 6: informed trading, part II.

5.5 Table 7: cross-sectional informed-trading tests

Read:

- The informed-trading effect is strongest for out-of-the-money options, consistent with leverage making OTM options attractive to informed traders.
- The effect is stronger on days before unexpected news events than on days without such news.

- It is also stronger for small option trades, consistent with informed traders splitting orders.
- Liquidity-demanding HFT remains positive and significant across the subsamples.

Economic message: The informed-trading proxy behaves as theory predicts. The HFT-spread link survives even in subsamples with lower informed-trading intensity.

Table 7
Testing the informed trading channel – cross-sectional analysis.

	$OPspread_{i,t}$ (i) - OTM	$OPspread_{i,t}$ (ii) - ITM	$OPspread_{i,t}$ (iii) - News	$OPspread_{i,t}$ (iv) - Nonews	$OPspread_{i,t}$ (v) - Small	$OPspread_{i,t}$ (vi) - Large
$Ins_{i,t-1}$	-0.05*** (-2.63)	-0.00 (-0.09)	-0.00 (-2.85)	-0.08*** (-2.21)	-0.04** (-2.51)	-0.01 (-1.26)
$Event_{i,t}$	-0.04* (-1.74)	-0.01 (-0.27)	-0.06* (-1.82)	0.00 (0.03)	-0.03 (-1.25)	-0.01 (-0.38)
$Ins_{i,t-1}Event_{i,t}$	0.05*** (3.03)	0.02 (0.74)	0.08** (2.53)	0.01 (0.43)	0.04** (2.35)	0.02* (1.71)
$SHFT_{i,t}^D$	0.04*** (3.59)	0.05*** (3.55)	0.03** (2.15)	0.04** (2.14)	0.03*** (2.89)	0.03*** (3.24)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Stock fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effect	No	No	No	No	No	No
N obs.	4623	4148	2100	2792	4892	4892
R ²	0.40	0.34	0.43	0.37	0.40	0.40

This table reports the results of testing the informed trading channel using the following model:
 $OPspread_{i,t} = \alpha_i + \gamma_1 Ins_{i,t-1} + \gamma_2 Event_{i,t} + \gamma_3 Ins_{i,t-1}Event_{i,t} + \gamma_4 SHFT_{i,t}^D + \sum_{k=1}^K \delta_k C_{k,i,t} + \epsilon_{i,t}$
where $OPspread_{i,t}$ is the proportional option bid-ask spread, $SHFT_{i,t}^D$ is the measure of HFTs' liquidity-demanding activities, and $Ins_{i,t}$ is the measure of informed trading. $Event_{i,t}$ is a dummy variable equal to one before October 16, 2009 (Rajaramam's arrest) and zero thereafter. The sample encompasses a period of two months before and after the event date, with the exclusion of the five trading days preceding the event date. $C_{k,i,t}$ is a set of k control variables, including variables from both the options and underlying markets. The options market variables are $Overvolume_{i,t}$, $Openbid_{i,t}$, $Openask_{i,t}$, $Openbid_{i,t} - Openask_{i,t}$, $Ogamma_{i,t}$, and $Overgamma_{i,t}$, and the stock market variables are $SPspread_{i,t}$ and $SVvolatility_{i,t}$. In columns 1 and 2, $Ins_{i,t}$ is calculated separately for out-of-the-money (OTM) and in-the-money (ITM) options. A put (call) is OTM if the stock price is above (below) the strike price. In columns 3 and 4, we conduct the regression analysis for unscheduled news (obtained from RavenPack database) and no-news days independently. In columns 5 and 6, $Ins_{i,t}$ is calculated separately for small and large trade sizes (based on the 100 contracts threshold). All models include firm fixed effects only. The standard errors used to compute the t-statistics (in brackets) are double clustered by firm and day. *, **, and *** denote the significance at 10%, 5%, and 1% respectively. For detailed definitions of variables refer to Table 1.

(i) and (ii) of Table 7 provide the results separately for OTM and ITM options, respectively. In columns (iii)-(vi), we provide the cross-sectional analysis presented in Table 7.

Table 7: informed-trading cross-sectional analysis.

5.6 Tables 8–9: 2SLS evidence

Read: Table 8:

- Fitted total HFT has a positive and significant effect on options spreads: 0.08.
- Fitted liquidity-supplying HFT is insignificant.
- Fitted liquidity-demanding HFT is positive and significant: 0.08.

Read: Table 9:

- In the IV latency-arbitrage specification, the interaction with *High* is positive and significant for both fitted total HFT and fitted demand HFT.
- This confirms that the effect is concentrated in stocks with more put–call parity violations.

Economic message: The IV results support a causal interpretation: aggressive HFT in the underlying stock increases options market-making costs, especially when latency-arbitrage opportunities are abundant.

Table 8
The impact of HFT activities on option spread – 2 SLS IV – second stage.

	$OPspread_{i,d}$ (i)	$OPspread_{i,d}$ (ii)	$OPspread_{i,d}$ (iii)
$\hat{SHFT}_{i,d}^{All}$	0.08** (2.03)		
$\hat{SHFT}_{i,d}^S$		0.01 (0.54)	
$\hat{SHFT}_{i,d}^D$			0.08** (2.02)
Controls	Yes	Yes	Yes
Stock fixed effect	Yes	Yes	Yes
Time fixed effect	No	No	No
N obs.	14,432	14,432	14,432
R ²	0.39	0.39	0.39

This table reports the results of estimating the impact of HFT in equity markets on the options bid-ask spread using the following models:

$SHFT_{i,d} = \alpha_i + \theta_1 IV_{i,d} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}$
 $OPspread_{i,d} = \alpha_i + \gamma_1 \hat{SHFT}_{i,d} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}$
where $OPspread_{i,d}$ is the proportional option bid-ask spread and $SHFT_{i,d}$ is one of three HFT measures ($\hat{SHFT}_{i,d}^{All}$, $\hat{SHFT}_{i,d}^S$ and $\hat{SHFT}_{i,d}^D$). $\hat{SHFT}_{i,d}$ is the fitted values of one of three HFT measures ($\hat{SHFT}_{i,d}^{All}$, $\hat{SHFT}_{i,d}^S$ and $\hat{SHFT}_{i,d}^D$) that is obtained by regressing the respective variables on $IV_{i,d}$, $IV_{i,d}$ is a dummy equal to one from June 5, 2009 to August 31, 2009, and zero for the other periods (from January 1, 2009 to June 4, 2009, and from September 1, 2009 to October 15, 2009). We exclude the 10 days before and after the event date. $C_{k,i,d}$ is a set of k control variables, including variables from both the options and underlying markets. The options market variables are $Ovolume_{i,d}$, $Oimplied_{i,d}$, $Oinpl_{i,d}$, $|Odelta_{i,d}|$, $Ogamma_{i,d}$, and $Ovega_{i,d}$, and the stock market variables are $SPspread_{i,d}$, and $SVolatility_{i,d}$. All models include firm fixed effect (α_i). The standard errors used to compute the t-statistics (in brackets) are double clustered by firm and day. *, **, and *** denote the significance at 10%, 5%, and 1% respectively. For detailed definitions of variables refer to Table 1.

(a) Table 8: 2SLS main effect.

Table 9
Testing the latency arbitrage channel – 2 SLS IV – second stage.

	$OPspread_{i,d}$ (i)	$OPspread_{i,d}$ (ii)
$\hat{SHFT}_{i,d}^{All}$	-0.32*** (-4.33)	
$\hat{SHFT}_{i,d}^D$		-0.29*** (-4.60)
$High_{i,d}$	-0.03** (-2.02)	-0.03** (-2.12)
$\hat{SHFT}_{i,d}^{All} High_{i,d}$	0.04*** (3.82)	
$\hat{SHFT}_{i,d}^D High_{i,d}$		0.05*** (3.69)
Controls	Yes	Yes
Stock fixed effect	Yes	Yes
Time fixed effect	No	No
N obs.	5899	5899
R ²	0.33	0.33

This table reports the results of testing the latency arbitrage channel using the following model:

$OPspread_{i,d} = \alpha_i + \gamma_1 \hat{SHFT}_{i,d} + \gamma_2 High_{i,d} + \gamma_3 \hat{SHFT}_{i,d} High_{i,d} + \sum_{k=1}^8 \delta_k C_{k,i,d} + \epsilon_{i,d}$
where $OPspread_{i,d}$ is the proportional option bid-ask spread and $\hat{SHFT}_{i,d}$ is the fitted values of one of two HFT measures ($\hat{SHFT}_{i,d}^{All}$ and $\hat{SHFT}_{i,d}^D$) that is obtained by regressing the respective variables on $IV_{i,d}$, $IV_{i,d}$ is a dummy equal to one from June 5, 2009 to August 31, 2009, and zero for the other periods (from January 1, 2009 to June 4, 2009, and from September 1, 2009 to October 15, 2009). We exclude the 10 days before and after the event date. $High_{i,d}$ is a dummy variable equal to one in day d if its $Npv_{i,d}$ (the number of put-call parity violations) value is above the median number of profitable put-call parity violations for all stocks. $C_{k,i,d}$ is a set of k control variables, including variables from both the options and underlying markets. The options market

(b) Table 9: 2SLS latency channel.

6 Short summary

- The paper studies whether stock-market HFT affects options-market liquidity.
- The main outcome is the proportional options bid-ask spread.
- The main HFT variables separate total, liquidity-supplying, and liquidity-demanding HFT.
- Total HFT is associated with wider options spreads.
- The effect is driven by liquidity-demanding HFT, not liquidity-supplying HFT.
- The paper proposes two mechanisms: latency arbitrage and informed trading.
- Latency arbitrage arises when HFTs exploit stale options quotes through put-call parity violations.
- The effect of HFT on options spreads is stronger for stocks with more profitable put-call parity violations.
- Informed trading in options raises adverse-selection costs and can generate temporary cross-market arbitrage opportunities.
- The Rajaratnam arrest is used as a shock to informed trading.
- Before the arrest, informed options trading predicts wider options spreads and more aggressive stock-market HFT.
- After controlling for informed trading, liquidity-demanding HFT remains significant.
- NASDAQ flash orders are used as an instrument for HFT activity.
- The 2SLS estimates confirm that aggressive HFT raises options-market spreads.
- The paper shows that microstructure effects can propagate across linked markets.

7 Conclusion

7.1 Core conclusion

Nimalendran, Rzayev, and Sagade show that HFT activity in the stock market can raise the cost of market making in the options market. The key effect is not passive HFT liquidity provision but aggressive, liquidity-demanding HFT.

7.2 Economic intuition

Options market makers quote prices that depend on the underlying stock price. When the stock price moves or when informed options trades reveal information, fast stock-market traders can exploit stale options quotes or related put–call parity violations. Anticipating this sniping risk and adverse selection, options market makers widen their bid–ask spreads.

7.3 Takeaway

The paper’s central lesson is cross-market: speed in one market can impose costs in another linked market. HFT may improve some dimensions of stock-market liquidity, but aggressive HFT can also increase options trading costs through latency arbitrage and informed-trading channels.